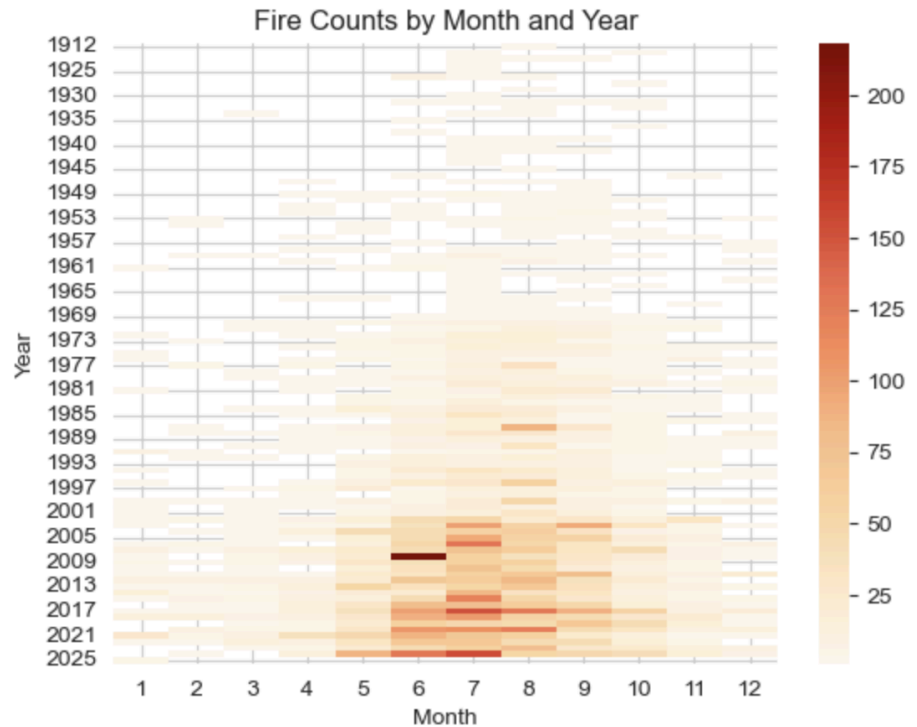


P4 Design Process

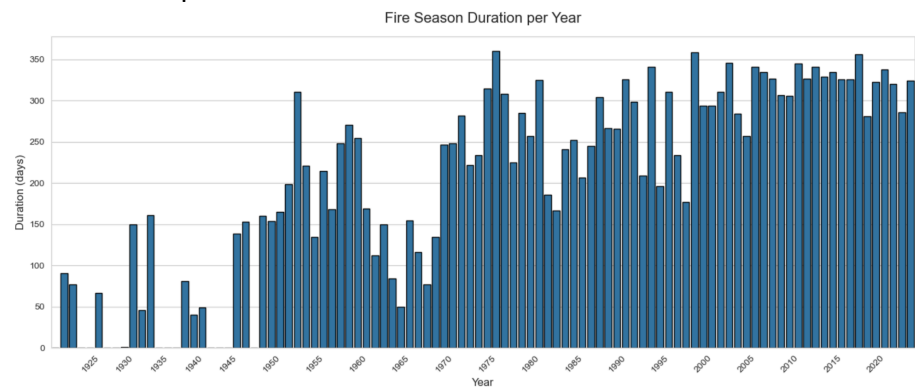
Due: EOD W10 Friday (12/06 @11:59 PM)

Use the following table to structure your notes about the design process you follow for Project 4. Feel free to expand/contract the size of the cells, and add/remove/change styles to your preference. Note that the template is currently structured to be consistent with the grading rubric, and includes spacing reflective of how much we expect you to write about each step of your design process (without including photos/screenshots). We discourage you from going substantially over/under these spacing guidelines—except in order to include screenshots of your process).

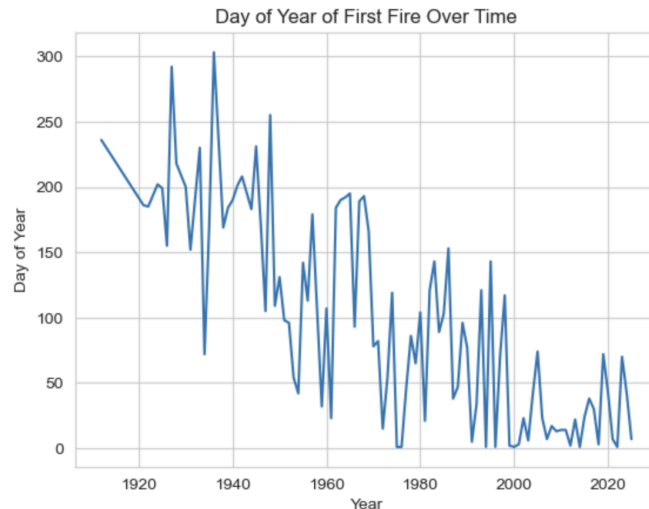
DATASET	
What dataset have you chosen, and why? How did you make this decision? Link: https://gis.data.cnra.ca.gov/datasets/CALFIRE-Forestry::california-fire-perimeters-all/about	We chose to use the California Fire perimeters (California Department of Forestry and Fire Protection) dataset because it seemed interesting and relevant, especially since we live in Southern California, where there are a lot of wildfires. The dataset had information about each fire's agency and responding unit, name, alarm date and containment date, cause, collection method, GIS calculated acres, and size, so there are plenty of variables to explore and a lot of directions we could go in for our data narrative. There were a couple of other datasets we were thinking about using but this one had a more comprehensive description of the data (descriptions of updates, criteria for entries to be added, updates on unit IDs, etc), which we thought would be helpful for this project. We wanted a dataset that allows us to tell meaningful stories and create clear visualizations, and this dataset fits those criteria.
DATA NARRATIVES	
PRIMARY NARRATIVE. What is the primary narrative you've chosen to tell? How did you decide on this narrative? (be sure to include photos/screenshots of any visualizations you create as you explore the dataset). Recall that a data narrative is a 'high level outline or 'topic' of a story; one that you will fill in with the specific details of the visualization(s) that you use to tell the story.	The data narrative we decided to tell was how the fire season in California has shifted since the 1900s. We decided on this by looking at our dataset to see what kind of information we had and what questions would be important to answer. We had some knowledge of a "fire season" and realized we could examine that with this dataset.



To illustrate possible trends, we graphed the fire count according to each year and month. This shows that the number of fires has been increasing and the months in which fires occur has been increasing (fire season has been getting longer). It is a bit harder to read, so it did not end up in our final version.

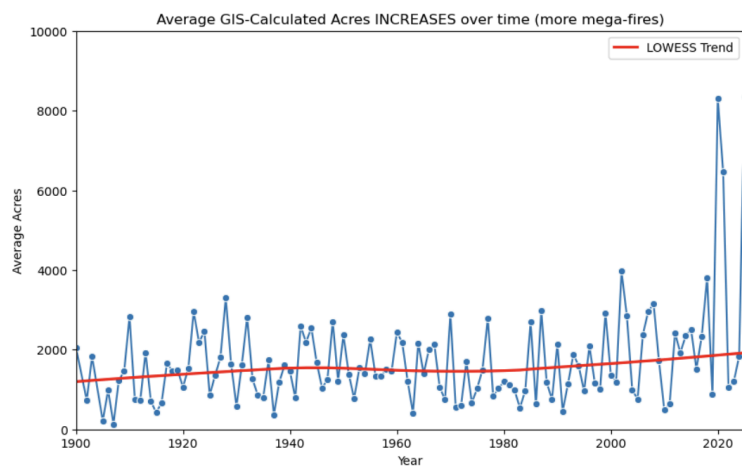


Looking at other graphs in news coverage about California wildfires, we decided to plot ones that were consistent with that for better readability/understanding.

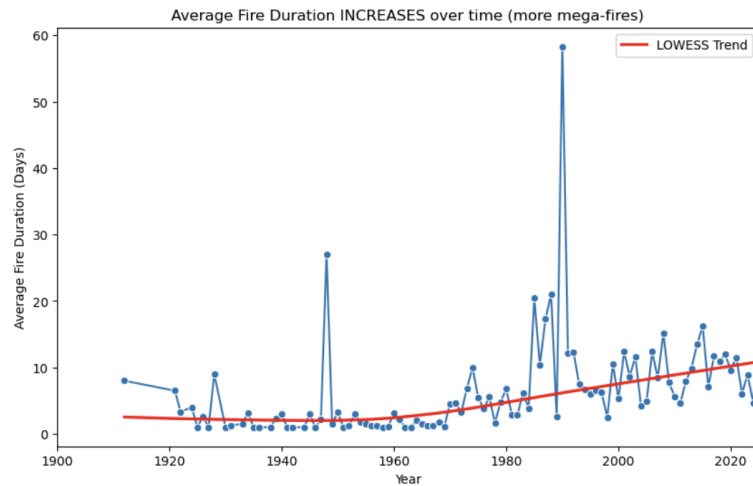


The other portion of the story was that alarm dates are getting earlier, so we plotted the first alarm date of each year.

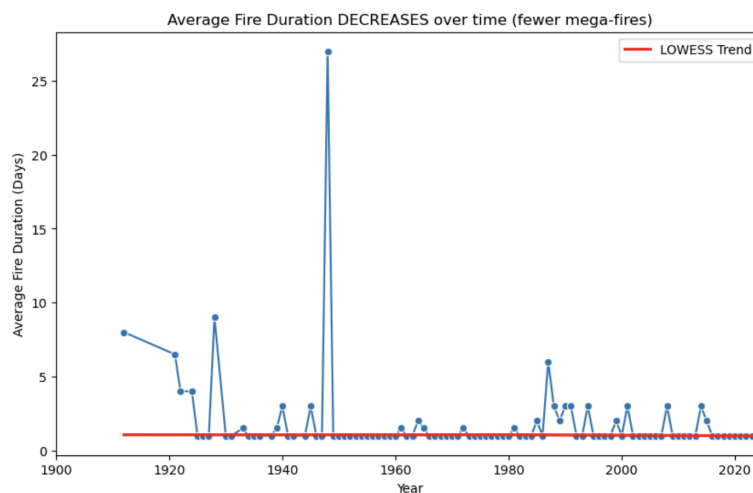
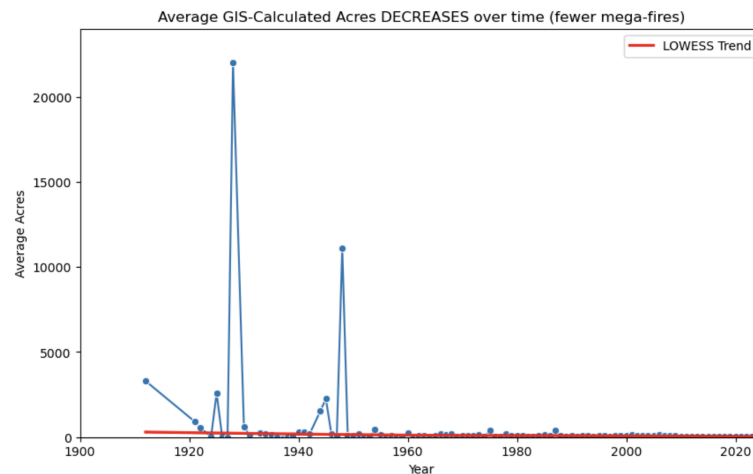
For the trickier portion of the project, we first looked at what the data actually said before manipulating the presentation.



We plotted the mean number of acres burned by wildfires over time to establish the “true” trends—the size has increased over time with mega-fires becoming more common and more extreme.



We also looked at using the median instead of mean to describe the average fire each year—since median ignores extreme outliers, we thought the visualizations might be deceptive.



We later explored other techniques for deception.

Contexts.

Who is your **intended audience**?

What is the **situational context** of their engagement with your data story?

What specific design choices will you make that will impact the audience's **trust** in your story?

(spacing developed for calibri text at font size 12; ~ 12 lines)

Audience: The intended audience is the general public (i.e. no requirement to know specifics of firefighting). Assumed that the audience is familiar with how to interpret common graphs, like bar charts or scatterplots.

Situational context: graph attached to a news article, expect readers to spend a little bit of time interpreting the graph (e.g. reading labels and legend, etc). No expectation of firefighting or terminology but assume knowledge of some common graph types.

Design choices: Use common graph types to make interpretation easier. Choose fonts and colors that fit a more educational or scientific look.

Outline.

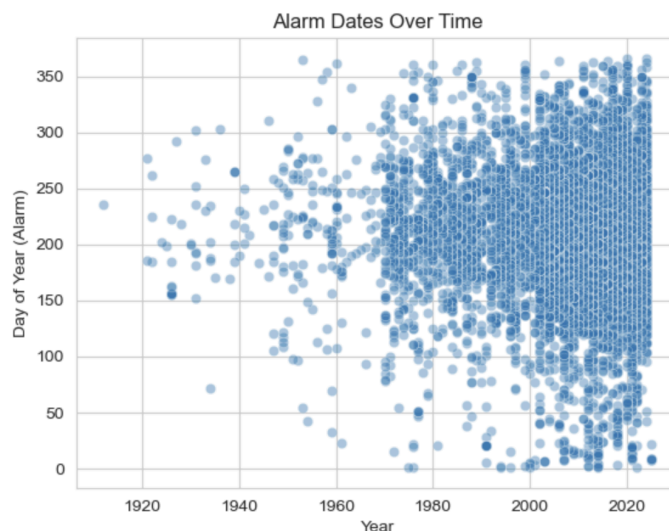
What is the (text) outline of the story? What are the main takeaway messages viewers should learn from their interaction with your story? You may want to include sketches and/or prototypes of the visualization(s) that will make up your story.

(spacing developed for calibri text at font size 12; ~ 16 lines)

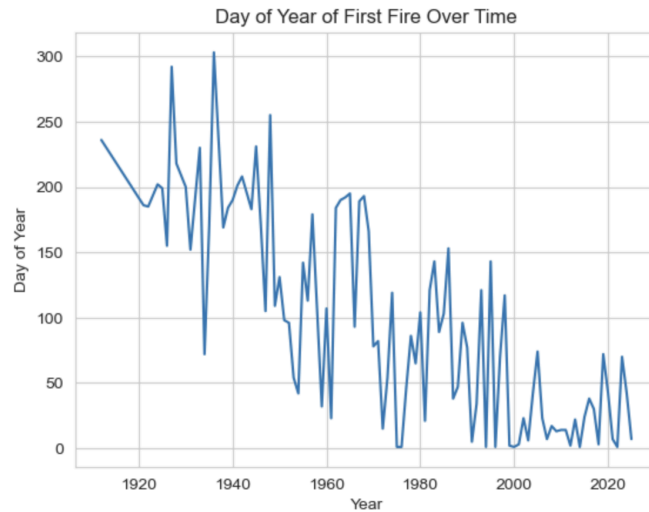
Story: We wanted to look at how the fire season in California has changed over time. Hypothesized that fires may be happening more frequently and most often in warmer, dryer months.

Main takeaway: Fire season is starting earlier and lasting longer. Historically, it has been concentrated to summer months (April-October), but has recently expanded to almost the entire year.

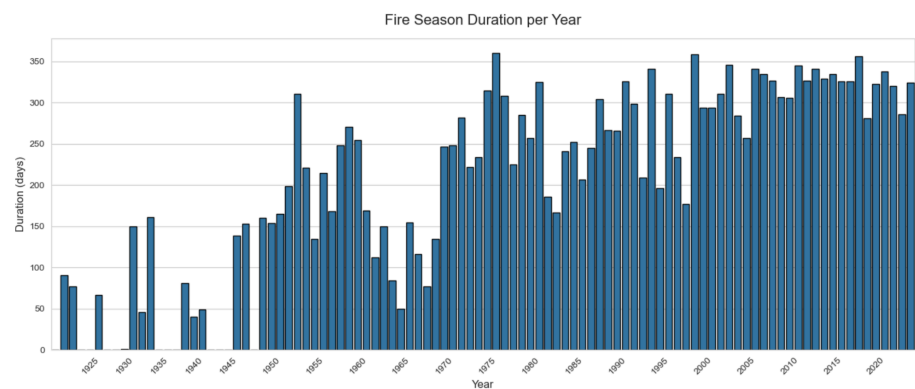
*captions for each graph



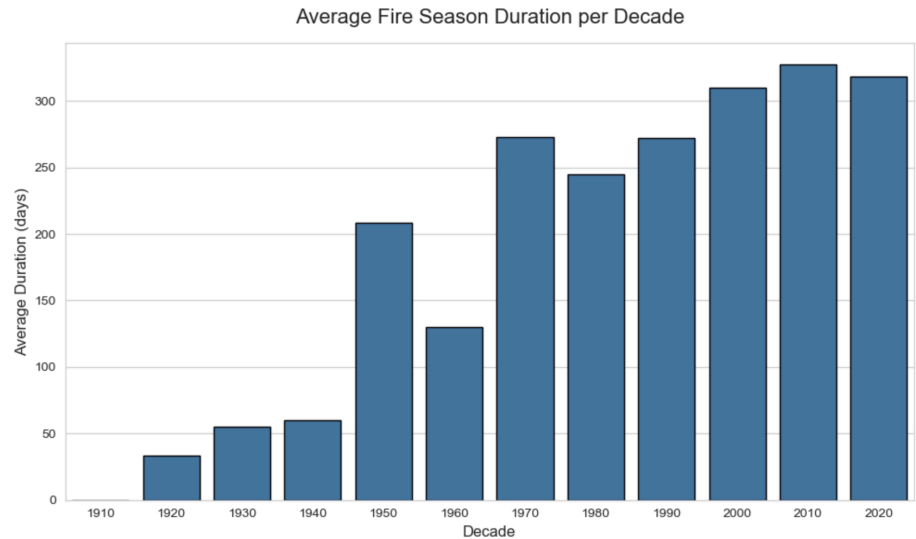
Number of fires per year is increasing. Fire season appears to be getting longer.



The first fire of the year is getting earlier. Used to be between 200-250th day (July-September), now is usually between 0-50.



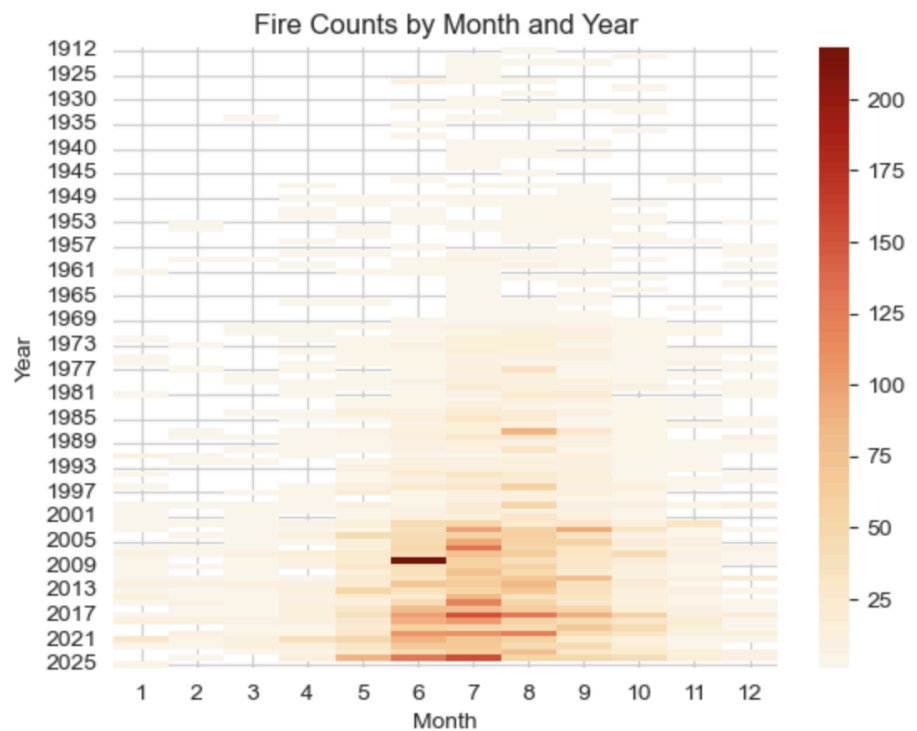
This bar plot shows the fire season duration for each year, and indicates that the length of the fire season is trending upwards. Fire season duration was calculated by (last alarm date of the year) – (first alarm date).



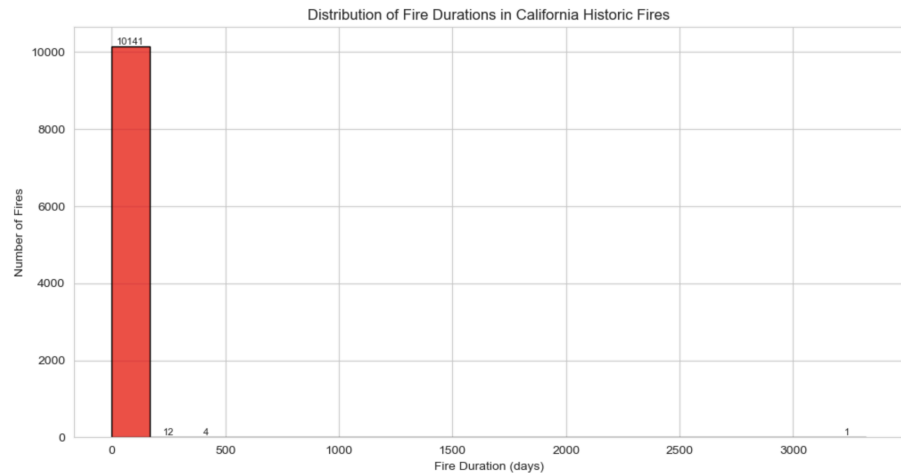
This is the same data, but broken up by 10 years. Each bar represents the average (mean) fire season duration for that decade. Shows a similar trend but looks less cluttered visually.

Work in Progress.

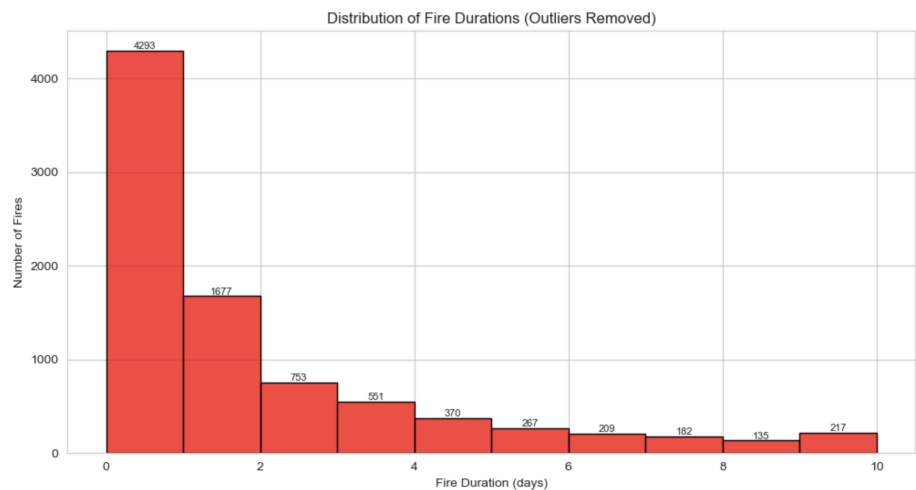
Include here images of any work-in-progress versions of your visualization(s) you think are notable, and want to discuss what you learned from them and/or why they did or did not make it into the final version.



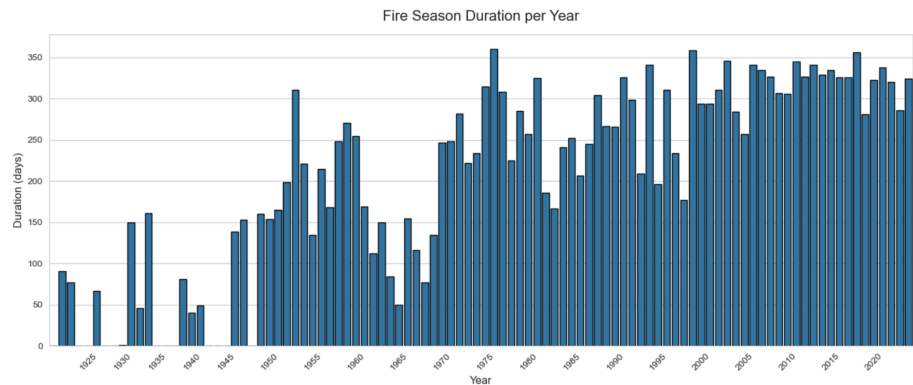
Concentration of fires that occurred each month and year. Uses color to encode the number of fires for a month/year combo. Note that this can provide a good overview, but is less specific because there are no labels on each grid square of the number of fires that happened that month.



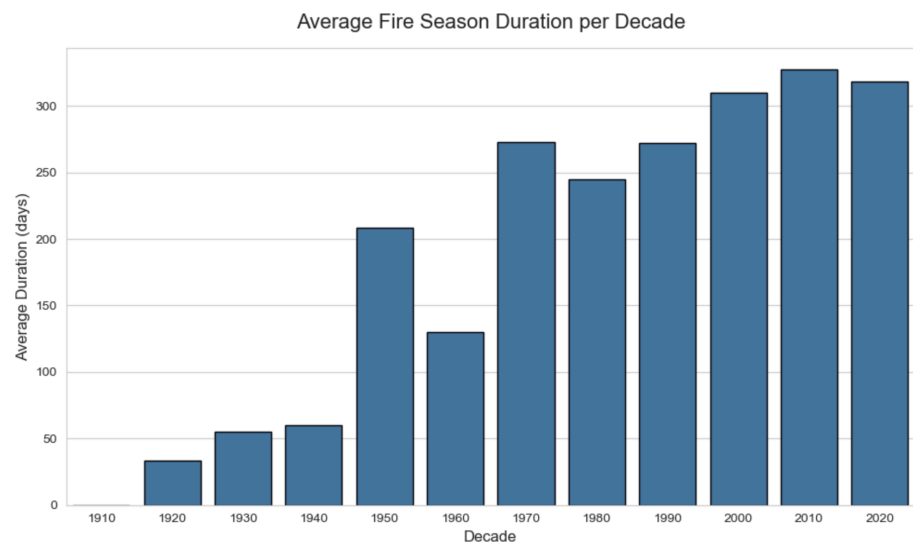
Fire duration had one very large outlier (3319 days to contain). Found datapoint and cross referenced with another source – ended up being inaccurate, was assuming it was misentered.



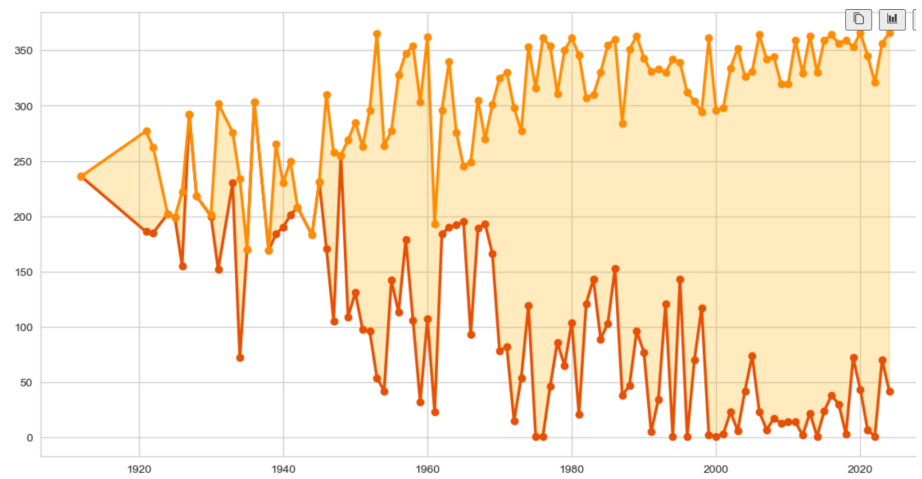
Replotted fire durations without outlier. Shows that most fires were contained within 1 day, has data points with containment up to 10 days. These bar plots were good, but we thought some of the other visualizations were more valuable.



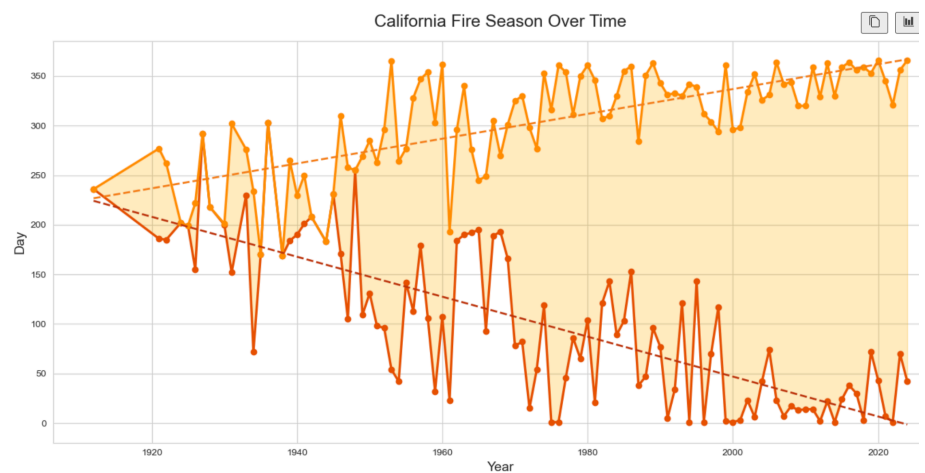
This bar plot shows the fire season duration for each year, and indicates that the length of the fire season is trending upwards. Fire season duration was calculated by (last alarm date of the year) – (first alarm date).



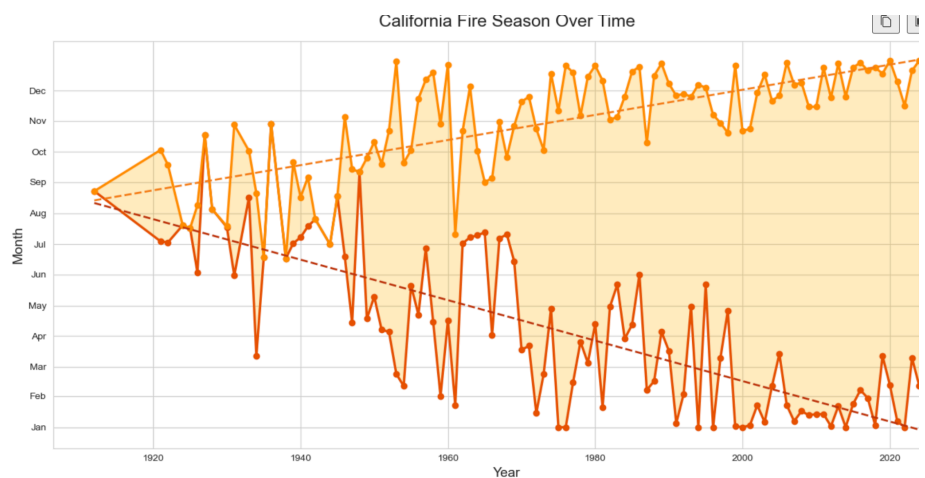
This is the same data, but broken up by 10 years. Each bar represents the average (mean) fire season duration for that decade. Shows a similar trend but looks less cluttered visually.

Main final visualization:

Initial plot of first alarm dates (red) and last alarm dates (orange) with a fill between them.

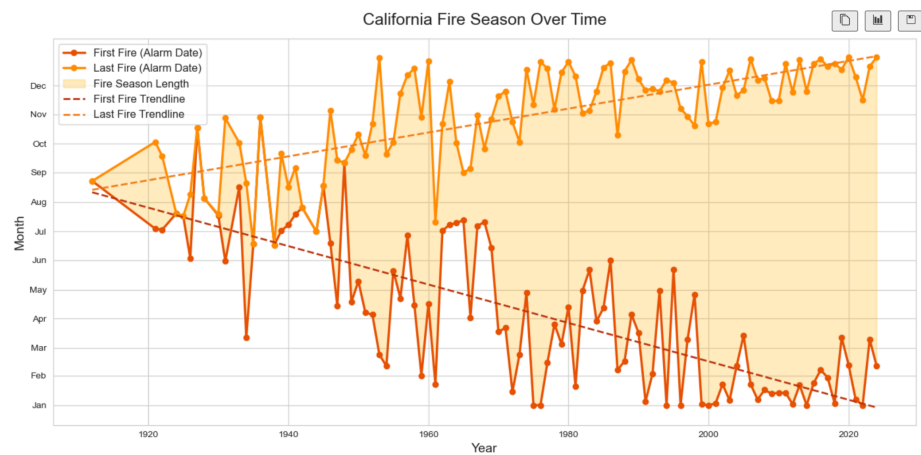


Add title, axes labels, trendlines for first and last fire alarm dates.



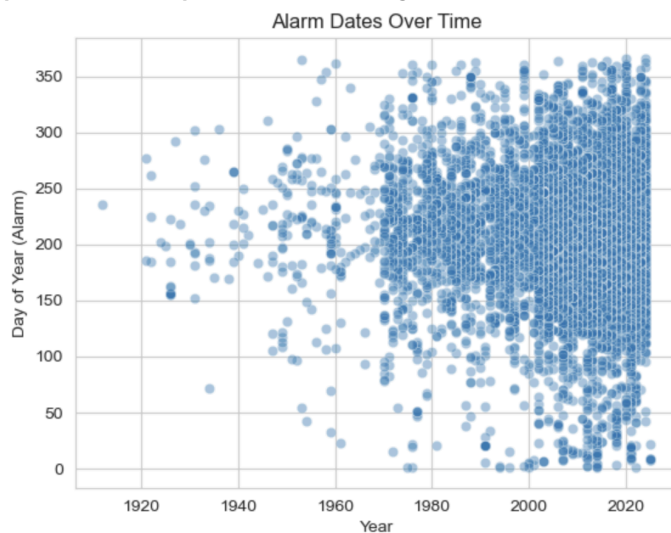
Change y axis label to month instead of day for improved

interpretability.

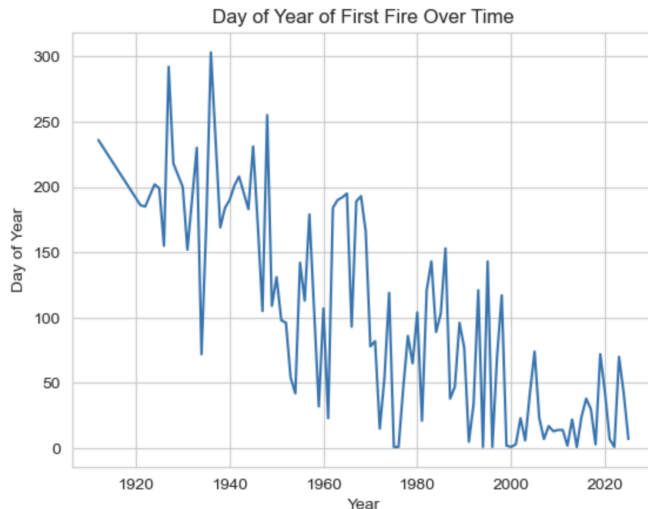


(final version w/ legend)

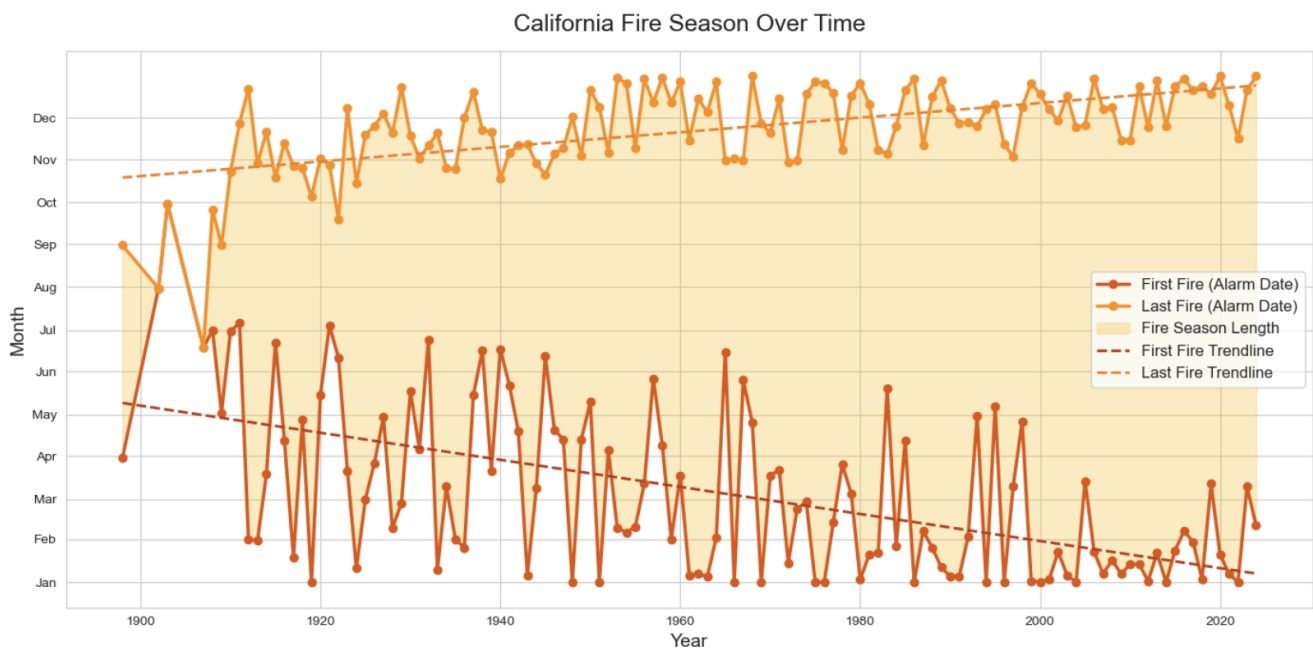
(FINAL VERSION) Include static image of each visualization in the final version of your data story)



The number of fires has increased over time. Concentrated mid-year when there are hotter and dryer conditions.



The date of the first fire of the year has been getting earlier over time.



Final graph that shows when the first and last alarm dates were for the year. Shaded area is the duration of the fire season. The orange dotted line is the trendline for the last fire of the year and the red dotted line is the trendline for the first fire of the year.

DATA STORY 2: For TRICKERY

Contexts.

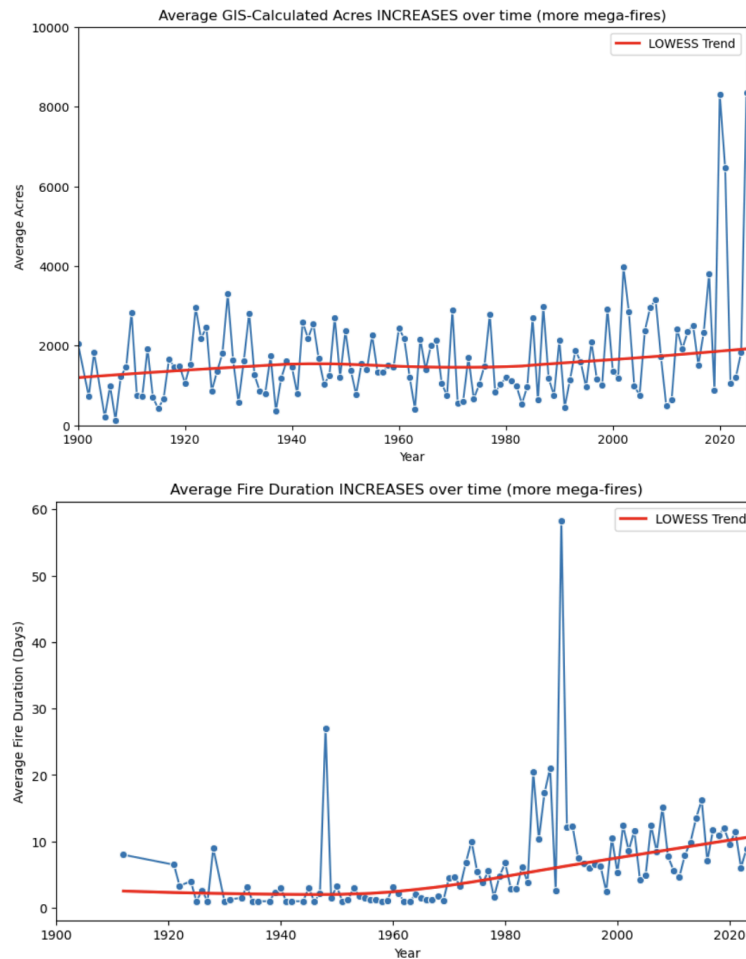
Who is your **intended audience**?

What is the **situational context** of their engagement with your data story?

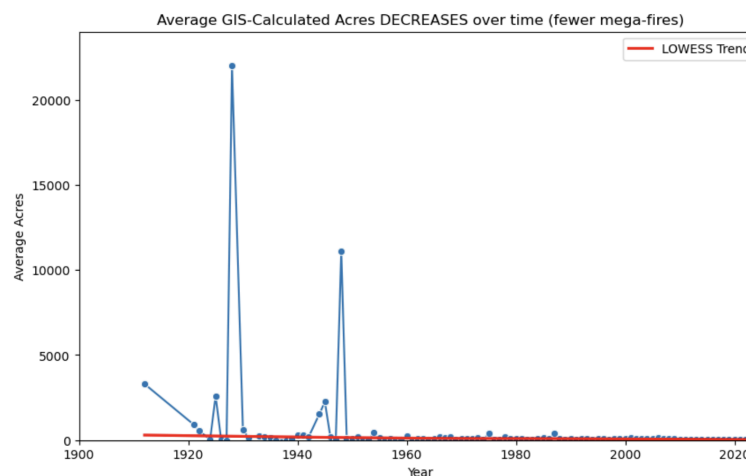
What specific design choices will you make that will impact the

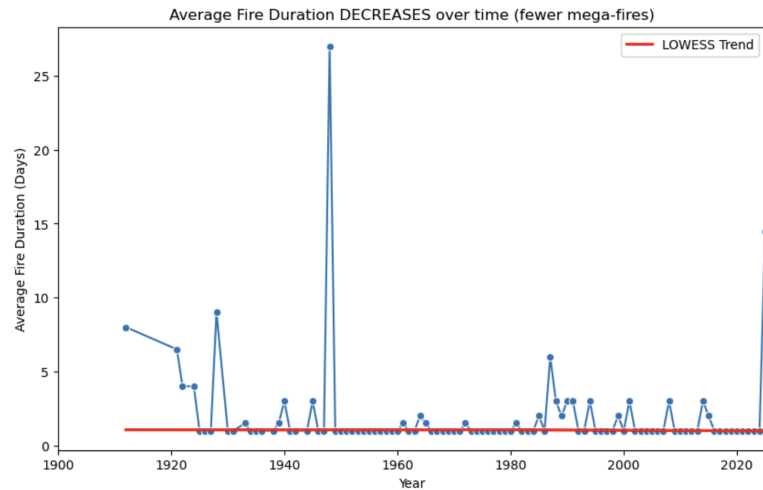
The intended audience is the general public (no assumption of knowledge of firefighting) who are familiar with reading common graph types like lineplots. The situational context is the plot is a news graphic, in something like a social media post, so it is intended to be glanced at briefly and conclusions reached quickly. Design choices we made to evoke trust are matching the aesthetics of the plot to other

audience's trust in your story?	graphs in news coverage about California wildfires—timeplots comparing across years are common, same color palette (white background with gray gridlines, lots of reds/oranges/yellows)—and transparency in data collection in the caption helps convince the reader the plot is trustworthy because readers likely assume since it is honest in one area, it will be honest overall.
Outline. What is the (text) outline of the story? What are the main takeaway messages viewers should learn from their interaction with your story? You may want to include sketches and/or prototypes of the visualization(s) that will make up your story. What specific design / encoding choices will you make in order to guide your readers to inappropriate or misleading takeaway messages?	The main takeaway should be that the duration of wildfires has actually been decreasing over time. Some of the design/encoding choices I made in order to deceive viewers were to invert the y-axis, which violates the convention of axes being the upper right of a cartesian plane with the assumption that higher on the plot means a larger y-value and an increase from left to right means an upward trend. I used a logarithmic scale instead of linear which exaggerates the changes from year to year and is not a common (or very interpretable) scale. Choosing to use only the mean duration each year makes it harder to interpret what is driving changes: are typical fires getting longer or are there more mega events (outliers) that pull the average while most fires remain the same duration? The red fill area is likely interpreted as additive rather than multiplicative, the viewer assuming the scale is linear rather than logarithmic so they might get the impression that the changes are greater than they really are. Also, the decision to use red as the color evokes a sense of danger and urgency, so the viewer might have a more emotional response to the graph than if it were just a black line on a white background, and therefore might not look at the details as closely or think about it as much.
Work in Progress. Include here images of any work-in-progress versions of your visualization(s) you think are notable, and want to discuss what you learned from them and/or why the did or did not make it into the final version.	Originally, we wanted to look at how the intensity of fires changed over time, measured by size and duration but later narrowed it to just duration (see below) Plots using mean as “average”—conclusion: size and duration tend to increase year over year. Increase in mega-fires contributes to this trend since outliers drag the mean in that direction (good, we want to take those fires into consideration)



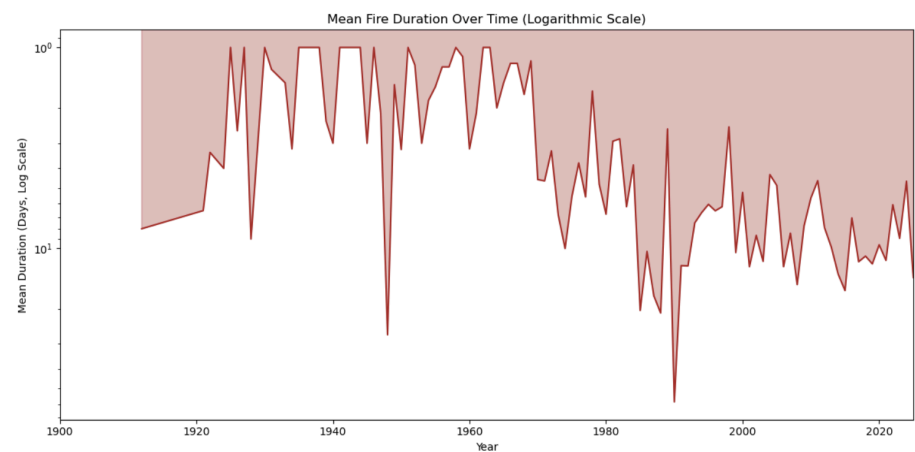
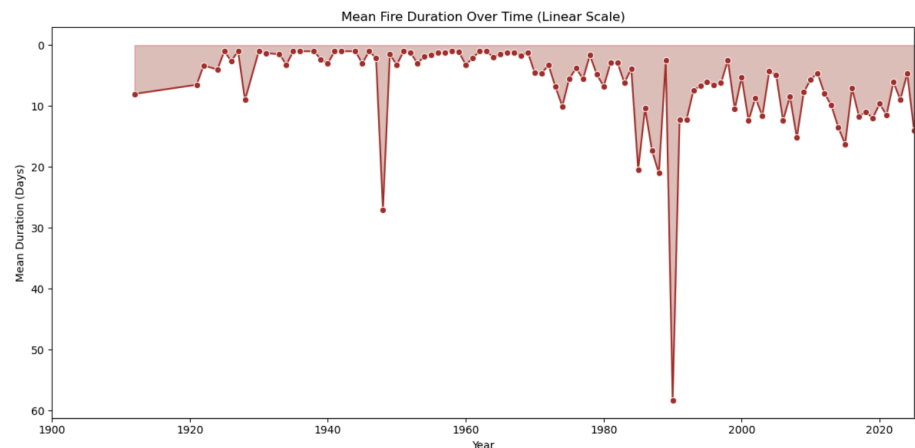
Plots using median as “average”—size and duration decrease year over year, 2025 is an outlier. Median does not take mega-fires into account, just modeling the “typical fire” which has not changed but there has been an increase in mega-fires that is not captured (bad, should not ignore)



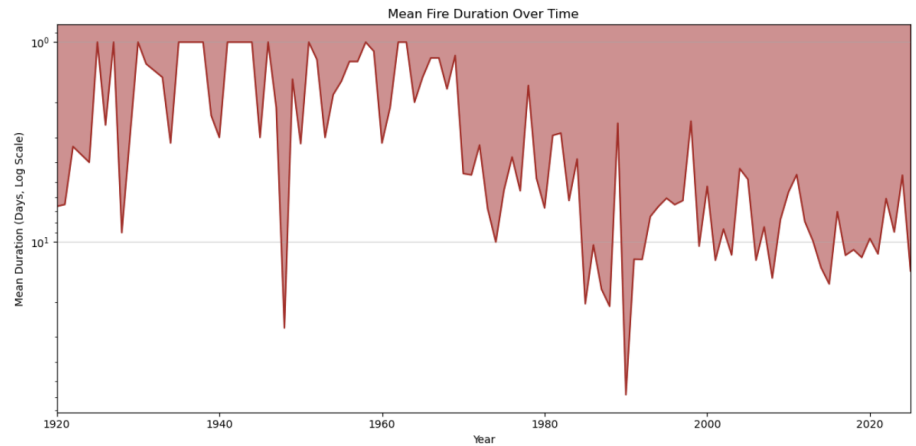


Plot with log scale to exaggerate changes in the trends, reversed y-axis violates assumption that graph sits in upper right of xy coordinate plane so conclusions are opposite of.

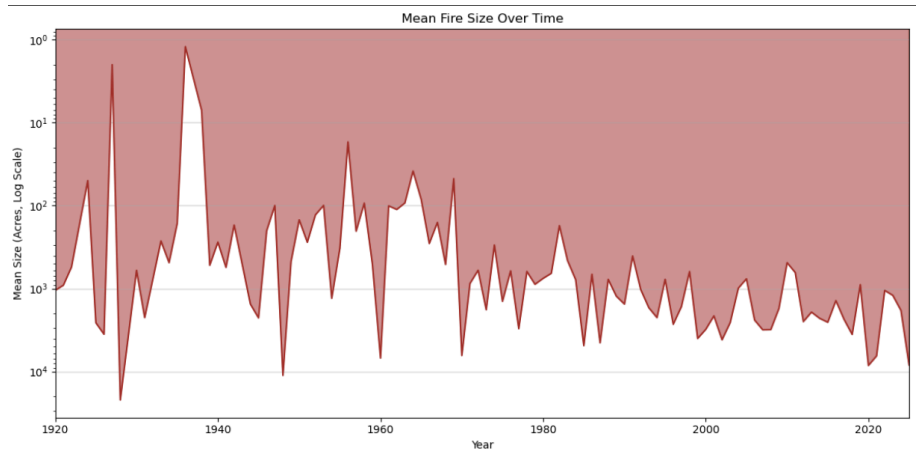
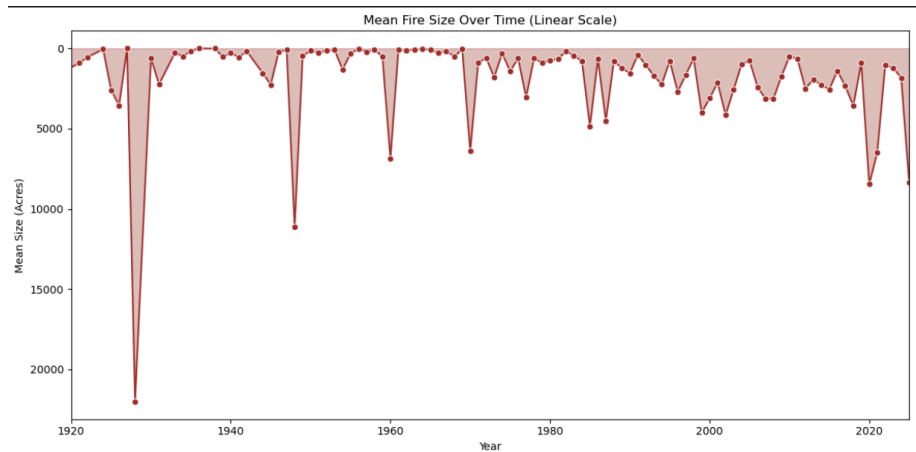
Mean Fire Duration:



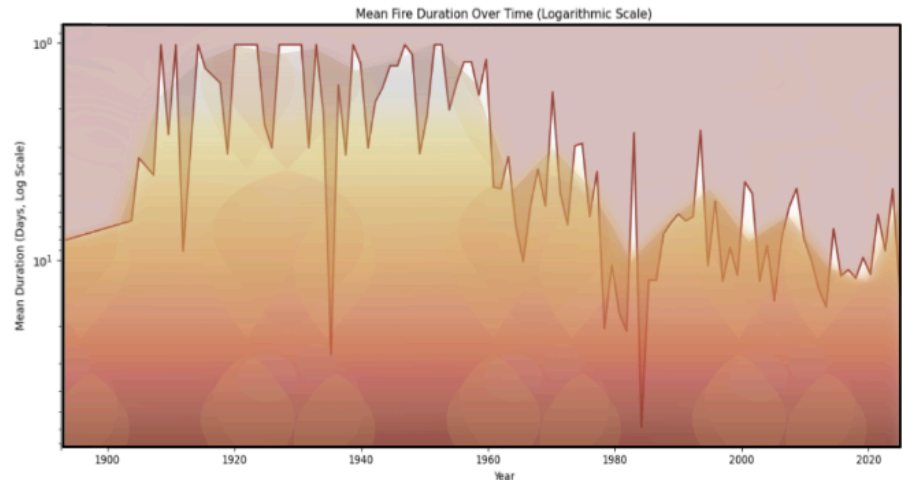
I decided to make the fill more opaque so the red was more intense in order to evoke a more visceral reaction



Mean fire size:

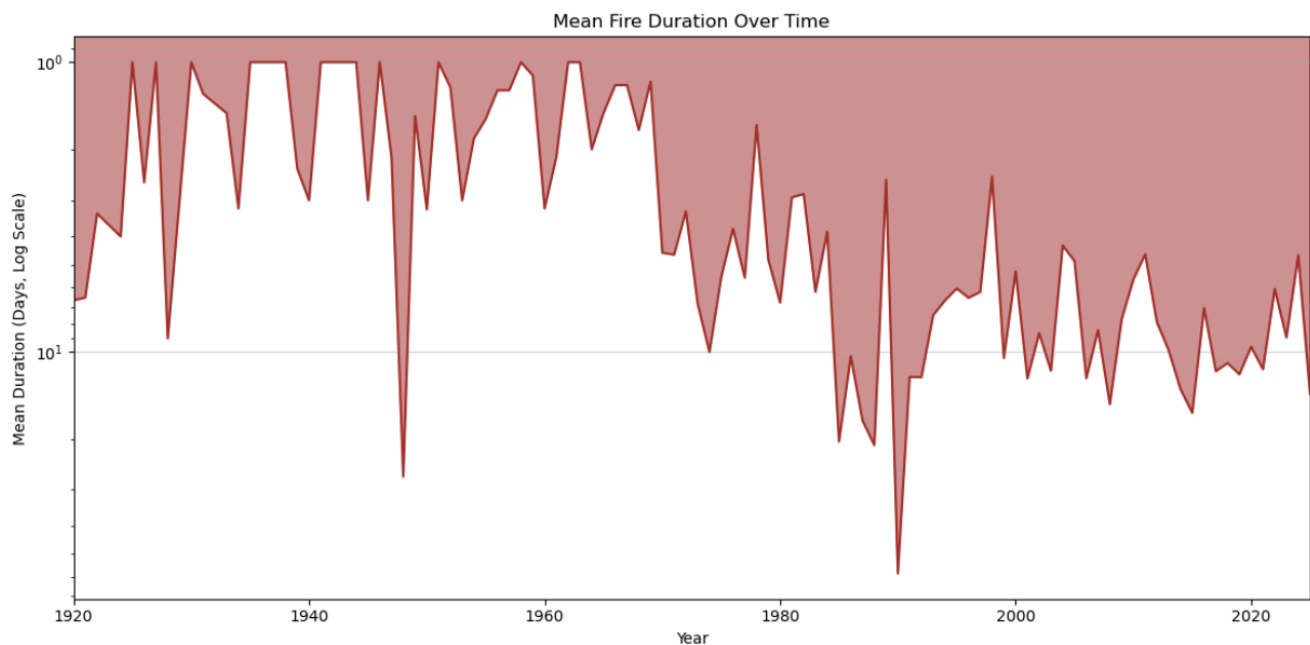


In both the duration and size plots, the log scale made the trends look more extreme, which we wanted but the size plot ended up looking like the acres burned stayed pretty constant. Therefore, for the final visualization, we decided to focus on duration only for a stronger conclusion/deception.



Add visual overlay to evoke the idea of flames but ultimately decided it was too much and did not look as trustworthy (did not add the visuals to the size plot).

(FINAL VERSION) Include static image of each visualization in the final version of your data story)



Average duration of California wildfires, 1920-present (early 2025)

Duration measured from alarm to containment date. Fires whose alarm and containment dates were the same were counted as one day. Values reported from Cal FIRE

DESIGN REFLECTION	
REFLECTING ON TRUST. What did you learn about how trust is built using design choices?	I learned that there are so many design choices that can be made, with either trust or deception in mind. I was much more aware of specific details, like axes, legends, word and color choices and how variables are encoded. I did not realize how many assumptions we make about graphs, like what the scale should be, the “correct” way to read something like a scatterplot, and how encodings like size are interpreted. I had not realized how much certain more aesthetic choices (ie font, colors, background, word choices) played into how trustworthy a graph is interpreted to be. It also emphasized the responsibility designers have to present information ethically, since even small choices can meaningfully shift a viewer’s perception.
REFLECTING ON TRICKERY. What did you learn about misleading with visualization design choices?	In this project, I learned just how easy it was to create a visualization that conveys the message you want, without changing the data itself. I learned to look at the axes much more closely—the scale of both axes (eg linear scale vs logarithmic scale) can dramatically change how the trend looks as well as flipping or compressing an axis to play off assumptions and conventions that an axis starts at 0 and increases outward (origin where the axes meet). Even a choice of mean vs median to describe the “average” affects what the trend looks like and can support entirely different data narratives depending on the underlying distribution. Aesthetics like color saturation, gridline strength, and filled areas that I would have said did not really make a difference can guide attention towards or away from certain aspects, leading you to a certain conclusion.